Fairness and Loan Thresholds

Randall Pruim

2026-06-29

Student Materials

- Worksheet
  - web
  - PDF

Overview

Ethics background required: Ideally, students should be familiar with the 4 ethical frameworks presented in the first year curriculum (virtue ethics, deontology, consequentialism, analogies) as that will provide them with a better context for discussion, but the activity could be done without that background.

Subject matter referred to in this lab: Fairness in machine learning algorithms (applied to a simplified loan threshold scenario).

Placement in overall ethics curriculum:

- Academic year: Year 2-4 core course that talks about machine learning, confusion matrices, or fairness.
- Recommended previous labs: First year ethics curriculum
- Recommended follow-up labs: No direct follow-up labs. But there is a similar interactive website at https://pair.withgoogle.com/explorables/measuring-fairness/ that could be used for further investigation (or as an alternative).

Time required:

- In class: 30 minutes or more.

Learning objectives:

- Students are invited to think about ways to quantify fairness.
- Students discover that there are multiple notions of fairness, and that they are not all compatible with each other.

- Students are encouraged to think ethically about fairness and how it plays out in situations that impact people.

Ethical issues to be considered: Fairness, diversity

Flow

Students work through a guided activity while reading through and interacting with the activity at https://web.archive.org/web/20260101103806/http://research.google.com/bigpicture/attacking-discrimination-in-ml/. (Note: This site is no longer available where it was originally posted, but the "Way Back Machine" provides a usable version of it.)

Instructors could modify the exercises or provide some additional introductory material if they think students will have a difficult time coming up with and understanding metrics for fairness. Alternatively, the first portion could be done as a class discussion to make sure everyone has a good understanding of the metrics before moving on to the interactive portion of the lab.

Preparation

- Read through the entire lab and use the sliders in the interactive activities.
- Students will need access to https://web.archive.org/web/20260101103806/http://research.google.com/bigpicture/attacking-discrimination-in-ml/. If working in groups, it is probably sufficient to have one computer per group.
- Print copies of the worksheet unless students are going to access this online.
- It may be useful to assign student roles: A driver to interact with the webiste, a recorder to take notes, a reporter to report back to the class, a facilitator to keep the group on task, etc. If there are more roles than students in a group, some students can have multiple roles.

Guide for Instructors

Lesson plan

Introduction (5 minutes)

Have students brainstorm types of automated decisions that could be based on assigning some sort of score to a person. (Examples: loans based on credit scores; car insurance rates based on driving score; college admissions based on GPA or test scores, or some score that combines these with other factors like extracurricular activities, community service, etc.; hiring based on some sort of score that combines education, experience, and other factors; sentencing based on a "likelihood of recidivism" score, etc.)

Point out that none of these sorts of scores are perfect – some wrong decisions will be made. Have students think about the types of wrong decisions that can be made (positive decision for someone who should not have been given a position decision, negative decision for someone who should not have been given a negative decision) and how various stakeholders are affected by the different types of errors.

Depending on how comfortable you think students will be coming up with metrics for fairness, either do the first portion of the activity as a class discussions or let students work in groups to come up with their own metrics.

Worksheet ($\geq$ 25 minutes)

At the point students should be ready to work through the worksheet. It may be useful to interact with groups as they are going along to make sure they are on the right track, are asking questions if they are confused, and are giving sufficiently thoughtful answers.

Reflection

If you have time, reflection can be done as a class discussion having the reporters from the groups share their groups' answers to some of the questions. If in-class time is short, some of the responses could be assigned as group homework. If you use a system like Perusall, you could have students post their answers to the questions and then have other students comment on them.

Additional resources

In this simple situation, the results of score-based decision-making can be summarised in a 2-by-2 table called a confusion matrix.
The Wikipeidia article on confusion matrices includes a large number of different metrics that can be applied to evaluate the performance of these systems.

A. Narayanan [1] provides an overview of 21 different fairness definitions (and their "politics"). This is a video recording of a tutorial given at ACM FAT* in 2018.

References

Bibliography

[1] A. Narayanan, "21 fairness definitions and their politics." [Online]. Available: https://youtu.be/jIXIuYdnyyk?si=U4LpUQ3wCrRjz5dA